

• COMPUTER VISION · REINFORCEMENT LEARNING · IMAGE MATCHING

Reinforcement Learning–Based Keypoint Selection for Efficient Image Matching

Smarter Keypoints | Faster Matching | Real-World Impact

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Why Keypoint Selection Matters

Too Many Keypoints

Standard detectors like ORB, BRISK, and SIFT generate an excessive number of keypoints per frame.

Many are Redundant

A significant portion of detected keypoints are noisy or uninformative, particularly in repetitive textures.

Higher Computational Cost

Oversized keypoint pools heavily bottleneck descriptor matching latency and RANSAC verification.

Not End-to-End Optimized

Classical detectors maximize local corner response rather than downstream geometric matching success.

CORE PROBLEM STATEMENT

How can we reduce the number of keypoints during image matching while strictly preserving geometric information?

OUR PROPOSED APPROACH

Instead of replacing existing detectors, insert a lightweight reinforcement learning (RL) selection layer to intelligently keep or reject keypoints.

A Lightweight RL-Based Keypoint Selection Pipeline

STAGE 01

Input Images

Extract candidate keypoints from input image pairs using ORB, BRISK, or SIFT detectors.

STAGE 02

State Representation

8D vector: normalized (x, y), scale, orientation, response, best/2nd-best descriptor distances, and ratio.

STAGE 03

RL Policy Network

Compact MLP ($8 \rightarrow 32 \rightarrow 16 \rightarrow 2$ with ReLU) outputting discrete Keep or Reject decisions.

STAGE 04

Verification

Pruned keypoint pool flows directly to descriptor matching followed by RANSAC geometric verification.

TRAINING OPTIMIZATION

REINFORCE algorithm targeting high matching quality with fewer keypoints

DETECTORS & ACTION

ORB / BRISK / SIFT • Keep vs Reject

FINAL TARGET

Geometrically Valid Matches

Dataset, Training Details & Evaluation

Benchmark & Detectors

HPatches Benchmark

116 sequences (580 image pairs): 57 viewpoint and 59 illumination sets for realistic evaluation under varied conditions.

Detectors Evaluated

ORB (binary), BRISK (binary), SIFT (floating-point).

Training Details

REINFORCE & Adam Optimizer

10,000 training episodes with an entropy weight of 0.05 to balance exploration and convergence stability.

Learning Rates

1×10^{-4} (ORB, BRISK) • 5×10^{-5} (SIFT)

Baselines & Metrics

Compared Methods

Full detection (all keypoints), Top-k (highest response), Random subsampling, and RL selection (Ours).

Evaluation Metrics

Keypoint reduction ratio, RANSAC inliers, MMA @ 1/3/5px, Wall-clock time.

Validation Goal: Quantify the reduction ratio while confirming preservation of RANSAC inliers across severe perspective and lighting changes.

Less Keypoints. More Meaningful Matches.

BRISK REDUCTION

90.67%

14,879.34 → 1,387.71 keypoints

SIFT REDUCTION

45.66%

9,668.77 → 5,254.01 keypoints

ORB REDUCTION

36.26%

998.47 → 636.43 keypoints

Key Takeaways

- **Substantial Pruning:** RL significantly cuts keypoints (up to 90.67% on BRISK) while maintaining useful matching performance, especially for ORB.
- **Observation:** Top-k remains competitive on BRISK metrics, and policy inference currently introduces wall-clock latency overhead.

Future Work & Conclusion

- **Next Steps:** Improve detector-specific reward functions, analyze episode training convergence, and optimize RL inference overhead.
- **Conclusion:** Demonstrates that a lightweight RL selection layer substantially cuts candidate keypoints, providing a practical step toward efficient image matching.